

Common questions

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6/1      | (a) | (i)  |  | award (1) for either of following<br>11429<br>11466                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | 1   |     | 1     | 1     | 1    |
|          |     | (ii) |  | award (1) for either of following<br><br>able to absorb <u>many times</u> their own mass (of water)<br>able to absorb <u>hundreds of times</u> their own mass (of water)<br><br>accept weight as alternative to mass<br><br>neutral answers able to absorb 11429 times their own mass (of water)<br>able to absorb more than their own mass (of water)<br>able to absorb lots of water                                                                                                                       |                 |     | 1   | 1     |       |      |
|          | (b) | (i)  |  | award (2) for all points plotted correctly – tolerance $\pm\frac{1}{2}$ square<br>award (1) for 4 or 5 points plotted correctly<br><br>award (1) for appropriate curve<br>do not accept point to point line                                                                                                                                                                                                                                                                                                  |                 | 3   |     | 3     | 3     | 3    |
|          |     | (ii) |  | award (1) for any of following<br>bead absorbs water at 40°C more quickly / at higher rate<br>bead absorbs more water at 40°C<br>bead absorbs water at 10°C more slowly / at lower rate<br>bead absorbs less water at 10°C<br><br>award (1) for any of following<br>bead becomes saturated (after 10 hours) in water at 40°C<br>bead stops absorbing water at 40°C (after 10 hours)<br>bead not yet saturated (after 10/12 hours) in water at 10°C<br>bead still absorbing water at 10°C (after 10/12 hours) |                 |     | 2   | 2     |       | 2    |
|          |     |      |  | Question 6/1 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0               | 4   | 3   | 7     | 4     | 6    |